

AfterShip: Turning to Google Cloud to support continued steep growth in e-commerce parcel tracking

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Using Google Cloud, AfterShip automates shipping processes and cuts costs while growing business that already ships 30 million e-commerce packages per month.

About AfterShip

Founded in 2011, [AfterShip](https://www.aftership.com/) (<https://www.aftership.com/>) provides e-commerce shipment, label and rate calculation and self-service return services to 300,000 merchants worldwide.

Industries: Technology

Location: Hong Kong

Google Cloud results

- Reduced deployment times from hours to minutes
- Gained ability to expand into new geographic markets quickly and cost-effectively
- Forecast lowering infrastructure costs by 30%

Deployment
versus
software
minutes

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Spanner (<https://cloud.google.com/spanner/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

Thirteen years' experience in e-commerce has given Teddy Chan, Chief Executive Officer and Chief Technology Officer, [AfterShip](https://www.aftership.com/) (<https://www.aftership.com/>), a deep understanding of the challenges of shipping and tracking packages to customers worldwide. "The key problem many merchants face is customers asking 'where is my order?' and 'when I will get the package?'" Chan says. "When I considered this issue, I came up with the idea of AfterShip." Chan helped found AfterShip in 2011 to enable merchants to keep track of packages sent to customers via a web portal or an

API. AfterShip also allows merchants to notify customers of anticipated delivery times.

“Using AfterShip, merchants can provide the same experience to buyers regardless of which couriers they use,” Chan says. “Merchants can also improve their customer engagement by including up-selling or marketing content with their delivery notifications.”

Hong Kong-headquartered AfterShip continues to grow quickly and now has a 40-person team. Thirty members of this team are based on the China special administrative region and 10 are based in India.

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*—Teddy Chan, Chief
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AfterShip*

By October 2017, the business was tracking about 30 million packages per month and had expanded its services to include label and rate calculation and self-service return. Revenue, package transaction numbers and team size have doubled every year for the past three years, while more than 300,000 merchants and 426 couriers are signed up to the service. Key customers include Wish, Etsy and Groupon. Close to half AfterShip’s customers are based in the United States, about one third in Europe and the remainder are located in Asia. AfterShip had initially delivered its applications and services from an incumbent public cloud service. However, the company wanted to continue its growth trajectory

while automating key infrastructure processes, implementing a continuous deployment model and controlling costs.

The business needed to achieve these objectives while maintaining a global presence and high-quality service. AfterShip started reviewing its options and decided to migrate to Google Cloud

(<https://cloud.google.com/>). “Google Cloud has a network of global datacentres with deep connectivity that enables us to place our infrastructure close to our customers,” Chan says.

“In addition, the ability to horizontally scale our global database using tools such as Google Cloud Spanner (<https://cloud.google.com/spanner/>) eliminates any limits to our geographic expansion.

Furthermore, the managed services provided through Google Cloud would allow us to focus on building better features for online merchants.” The reliability provided by Google Cloud would also enable AfterShip to meet the stringent service level requirements of large digital marketplaces in the United States, Asia and elsewhere.

“Google Cloud could manage the high volumes and enable us to deliver the service levels that would realise our ambition of becoming the number one tracking API platform in the world,” Chan says. “For example, with Google Cloud, we can provide a 99.95% monthly uptime service level to our customers.” Finally, Google Cloud provided managed solutions, including Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) powered by open source container orchestrator Kubernetes, that would enable AfterShip to automate processes

such as scaling and enable its team to focus on developing applications.

AfterShip has moved its websites into Google Cloud infrastructure in three datacentres around the world and anticipates completing the migration in Q4 2017. “Google provided a lot of assistance, particularly early in the project when we needed it,” Chan says. “They briefed us on several services we hadn’t known about that could replace the equivalents in the public cloud we were using previously.” The business then completed the migration using its own skilled team members. As well as Google Kubernetes Engine and Google Cloud Spanner, AfterShip is using Google BigQuery (<https://cloud.google.com/bigquery/>) to store and analyse transaction information.

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Deployment times down from one hour to two minutes

With deployment times falling from up to one hour in its previous cloud environment to about two minutes in Google Cloud, AfterShip has been able to adopt a continuous deployment model. “This has improved our service levels,” Chan says. “If there are any issues we can fix them quickly, while we can iterate faster to create new features in response to customer requests or changes in the market. “This enables us to continue to lead our competitors.”

Targeting a 30 percent reduction in costs

AfterShip is now targeting a 30% reduction in costs by optimising its use of Docker containerisation technology on Google Cloud.

“By using Docker with Kubernetes, we have been able to fine-tune our use of compute resources and better control our costs,” Chan says. “We’re extremely pleased with Google Cloud as it really is built for engineers,” he adds. “In addition, its documentation is extremely clear, allowing us to troubleshoot or carry out activities on the platform ourselves. “We look forward to continuing to grow and extend our package tracking and associated services with Google Cloud.”